

Appendix: Choosing Better NLDL Layouts by Evaluating the Model in the High-dimensional Data Space

Jayani P. Gamage

Econometrics & Business Statistics, Monash University
and

Dianne Cook

Econometrics & Business Statistics, Monash University
and

Paul Harrison

MGBP, BDInstitute, Monash University
and

Michael Lydeamore

Econometrics & Business Statistics, Monash University
and

Thiyanga S. Talagala

Statistics, University of Sri Jayewardenepura

April 29, 2026

1 Methods and hyper-parameters used to generate layouts

Table 1 contains the list of methods and hyper-parameters used for each of the layouts shown in the paper.

Figure	NLDR method	Hyper-parameter(s)
1a	UMAP	n_neighbors = 30, min_dist = 0.3
1b	UMAP	n_neighbors = 5, min_dist = 0.8
1c	UMAP	n_neighbors = 5, min_dist = 0.01
1d	tSNE	perplexity = 5
1e	tSNE	perplexity = 30
1f	PHATE	knn = 5
1g	TriMAP	n_inliers = 12, n_outliers = 4, n_random = 3
1h	PaCMAP	n_neighbors = 30, init = random, MN_ratio = 0.9, FP_ratio = 5
2	tSNE	perplexity = 47
4a	tSNE	perplexity = 47
5b	tSNE	perplexity = 47
6	tSNE	perplexity = 47
8a	tSNE	perplexity = 47
8b	tSNE	perplexity = 62
8c	UMAP	n_neighbors = 15, min_dist = 0.1
8d	PHATE	knn = 5

(continued)

Figure	NLDR method	Hyper-parameter(s)
8e	TriMAP	n_inliers = 12, n_outliers = 4, n_random = 3
8f	PaCMAP	n_neighbors = 10, init = random, MN_ratio = 0.5, FP_ratio = 2
10a	UMAP	n_neighbors = 30, min_dist = 0.3
10b	UMAP	n_neighbors = 5, min_dist = 0.8
10c	UMAP	n_neighbors = 5, min_dist = 0.01
10d	tSNE	perplexity = 5
10e	tSNE	perplexity = 30
10f	PHATE	knn = 5
10g	TriMAP	n_inliers = 12, n_outliers = 4, n_random = 3
10h	PaCMAP	n_neighbors = 30, init = random, MN_ratio = 0.9, FP_ratio = 5
11a	UMAP	n_neighbors = 30, min_dist = 0.3
11e	tSNE	perplexity = 30
12a	tSNE	perplexity = 30
12b	tSNE	perplexity = 89
12c	UMAP	n_neighbors = 15, min_dist = 0.1
12d	PHATE	knn = 5
12e	TriMAP	n_inliers = 12, n_outliers = 4, n_random = 3
12f	PaCMAP	n_neighbors = 10, init = random, MN_ratio = 0.5, FP_ratio = 2

(continued)

Figure	NLDR method	Hyper-parameter(s)
13a	tSNE	perplexity = 30
14a	tSNE	perplexity = 30
A4a	tSNE	perplexity = 71
A4b	UMAP	n_neighbors = 15, min_dist = 0.1
A4c	PaCMAP	n_neighbors = 10, init = random, MN_ratio = 0.5, FP_ratio = 2
A5	tSNE	perplexity = 52
A6a	UMAP	n_neighbors = 30, min_dist = 0.3
A6b	tSNE	perplexity = 30
A7a	UMAP	n_neighbors = 30, min_dist = 0.3
A7b	tSNE	perplexity = 30
A8a	UMAP	n_neighbors = 30, min_dist = 0.3
A8b	tSNE	perplexity = 30
A9a	UMAP	n_neighbors = 30, min_dist = 0.3
A9b	UMAP	n_neighbors = 5, min_dist = 0.8
A9c	UMAP	n_neighbors = 5, min_dist = 0.01
A9d	tSNE	perplexity = 5
A9e	tSNE	perplexity = 30
A9f	PHATE	knn = 5
A9g	TriMAP	n_inliers = 12, n_outliers = 4, n_random = 3

(continued)

Figure	NLDR method	Hyper-parameter(s)
A9h	PaCMAP	n_neighbors = 30, init = random, MN_ratio = 0.9, FP_ratio = 5
A10a	tSNE	perplexity = 30
A10b	tSNE	perplexity = 89
A10c	UMAP	n_neighbors = 15, min_dist = 0.1
A10d	PHATE	knn = 5
A10e	TriMAP	n_inliers = 12, n_outliers = 4, n_random = 3
A10f	PaCMAP	n_neighbors = 10, init = random, MN_ratio = 0.5, FP_ratio = 2

Table 1: NLDR methods and hyper-parameters used for each Figure in the main paper.

2 Videos links

Animations of the p - D tours that produced specific projections shown in some figures in the main paper are available on YouTube at the links given in Table 2.

Figure	URL
4	youtu.be/yHKTHK4UBiU
5	youtu.be/Fukiminr090
11	youtu.be/3VfK3M2gnZM , youtu.be/Es84bwQcndU
13	youtu.be/sUcGd57Swdg , youtu.be/QiklCjELUxo

Table 2: Videos of the langevitour animations and the linked plots.

3 Notation

Notation	Description
n, p, k	number of observations, variables, embedding dimension, respectively
$\mathbf{X}$	$n \times p$ -dimensional data
$\mathbf{x}$	p -dimensional observation ($1 \times p$)
$\mathbf{x}^*$	new p -dimensional observation ($1 \times p$)
$\hat{\mathbf{X}}$	$n \times p$ -dimensional fitted values
$\mathbf{Y}$	$n \times k$ -dimensional data (layout)
$\mathbf{y}$	k -dimensional observation ($1 \times k$)
$\hat{\mathbf{Y}}$	$n \times k$ -dimensional fitted values
$\mathbf{y}^*$	new k -dimensional observation ($1 \times k$)
e_i	Residual of observation i , computed on p -dimensional data
P	orthonormal basis, generating a d -dimensional linear projection of p -dimensional data
T	true model
g	functional mapping from p -D to k -D, especially as prescribed by NLDR
θ	(Hyper-) parameters for NLDR method
r	ranges of the embedding components
$C^{(j)}$	j -dimensional bin centers
(b_1, b_2)	number of bins in each direction
(a_1, a_2)	binwidths, distance between centroids in each direction
(s_1, s_2)	starting coordinates of the hexagonal grid
q	buffer to ensure hexgrid covers data, proportion of data range, 0-1
m	number of non-empty bins
b	number of hexagons in the grid
h	hexagonal id
H_h	set of observations in bin h
l	side length
A	area
n_h	number of points in hexagon h (bin count)
w_h	standardized number of points in hexagon h (standardized bin counts)
g	mapping from p to k dimensions
$\tilde{g}$	pseudo-inverse of g
$\hat{g}$	estimate of g

Table 3: Summary of notation for describing new methodology.

4 Scripts

Folder	Script	Description
script	additional_functions.R	Helper functions to render the main paper.
script	evaluation.py	Python script implementing additional evaluation metrics such as RTA and GS.
script	nldr_code.R	Wrapper functions for running multiple NLDR methods (UMAP, tSNE, PHATE, PaCMAP, TriMAP) with different parameters.
two_nonlinear	01_gen_data.R	Generates the 2NC7 dataset.

(continued)

Folder	Script	Description
two_nonlinear	02_gen_true_model.R	Creates the true structure of 2NC7 data.
two_nonlinear	03_gen_embeddings.R	Computes multiple NLDR embeddings for the 2NC7 data.
two_nonlinear	04_gen_mse_for_diff_methods.R	Computes HBE with varying bin widths (a_1) for all NLDR embeddings.
two_nonlinear	05_gen_rm_lwd_mse.R	Computes HBE with varying low density bin cutoff for all three binwidth (a_1) choices.
two_nonlinear	06_gen_model_with_tSNE.R	Fits the model for the layout a.

(continued)

Folder	Script	Description
two_nonlinear	07_example_evaluation_metrics.R	Calculates evaluation metrics for all NLDR layouts.
two_nonlinear	08_gen_model_with_PHATE.R	Fits the model for the layout c.
five_gau_clusters	01_five_gaussian_cluster_data_emb.R	Generates data and multiple NLDR embeddings.
five_gau_clusters	02_gen_model_with_tSNE.R	Fits the model for the layout a.
five_gau_clusters	03_gen_model_with_UMAP.R	Fits the model for the layout b.
five_gau_clusters	04_gen_model_with_PaCMAP.R	Fits the model for the layout c.
c_shaped_dens_str	01_gen_data.R	Generates the 2- <i>D</i> curved sheet dataset.
c_shaped_dens_str	02_gen_embeddings_uni_dens.R	Generates multiple NLDR embeddings.

(continued)

Folder	Script	Description
c_shaped_dens_str	03_gen_model_with_tSNE.R	Fits the model for the tSNE layout.
pbmc3k	01_obtain_pca_author.R	Obtains authors' PCA results.
pbmc3k	02_obtain_umap_authors.R	Obtains authors' UMAP embeddings.
pbmc3k	03_gen_umap_diff_param.R	Generates multiple UMAP embeddings with different hyperparameter values.
pbmc3k	04_gen_tsne_diff_param.R	Generates multiple tSNE embeddings with different hyperparameter values.

(continued)

Folder	Script	Description
pbmc3k	05_gen_phate.R	Generates a PHATE embeddings with default hyper-parameters.
pbmc3k	06_gen_trimap.R	Generates a TriMAP embeddings with default hyper-parameters.
pbmc3k	07_gen_pacmap.R	Generates a PaCMAP embeddings with default hyper-parameters.
pbmc3k	08_gen_mse_for_diff_methods.R	Computes HBE with varying bin widths (a_1) for all NLDR embeddings.

(continued)

Folder	Script	Description
pbmc3k	09_gen_scDEED.R	Generates UMAP embeddings from scDEED results.
pbmc3k	10_pre_process_for_embedding.R	Generates PBMC3k data used for scDEED results.
pbmc3k	11_gen_mse_for_diff_tsne_scD.R	Computes HBE with varying bin widths (a_1) for tSNE embeddings.
pbmc3k	12_gen_mse_for_diff_umap_scD.R	Computes HBE with varying bin widths (a_1) for UMAP embeddings.
pbmc3k	13_gen_model_with_UMAP.R	Fits the model for the layout a.
pbmc3k	14_gen_model_with_tSNE.R	Fits the model for the layout e.

(continued)

Folder	Script	Description
pbmc3k	15_gen_model_with_UMAP_scD.R	Fits the model for the layout a.
pbmc3k	16_gen_model_with_tSNE_scD.R	Fits the model for the layout b.
pbmc3k	17_evaluation_metrics.R	Calculates evaluation metrics for all NLDR layouts.
pbmc3k	18_evaluation_metrics_scD.R	Calculates evaluation metrics for all NLDR layouts.
mnist	01_data_preprocessing.R	Computes first 10 principal components and save data.
mnist	02_gen_diff_embeddings.R	Generates multiple NLDR embeddings.

(continued)

Folder	Script	Description
mnist	03_gen_mse_for_diff_methods.R	Computes HBE with varying bin widths (a_1) for all NLDR embeddings.
mnist	04_gen_model_with_tSNE.R	Fits the model for the layout a.
mnist	05_evaluation_metrics.R	Calculates evaluation metrics for all NLDR layouts.
mnist	06_link_brush_layout_e.R	Creates interactive linked brushing with layout e.

Table 4: R and Python script files used to generate outputs in the main paper.

5 Generating the 2NC7 data

This data is constructed by simulating two clusters, each consisting of 1000 observations. The C-shaped cluster is generated from $\theta \sim U(-3\pi/2, 0)$, $X_1 = \sin(\theta)$, $X_2 \sim U(0, 2)$ (adding thickness to the C), $X_3 = \text{sign}(\theta) \times (\cos(\theta) - 1)$, $X_4 = \cos(\theta)$. Observations lie on a 2- D

manifold in 7- D . The other cluster is from $X_1 \sim U(0, 2)$, $X_2 \sim U(0, 3)$, $X_3 = -(X_1^3 + X_2)$, and $X_4 \sim U(0, 2)$. It is also curved, but observations lie on a 3- D manifold in 7- D . Three more variables, X_5, X_6, X_7 , that are small amounts of pure noise are added. We would consider $T = (X_1, X_2, X_3, X_4)$ to be the geometric structure (true model) that we hope to capture.

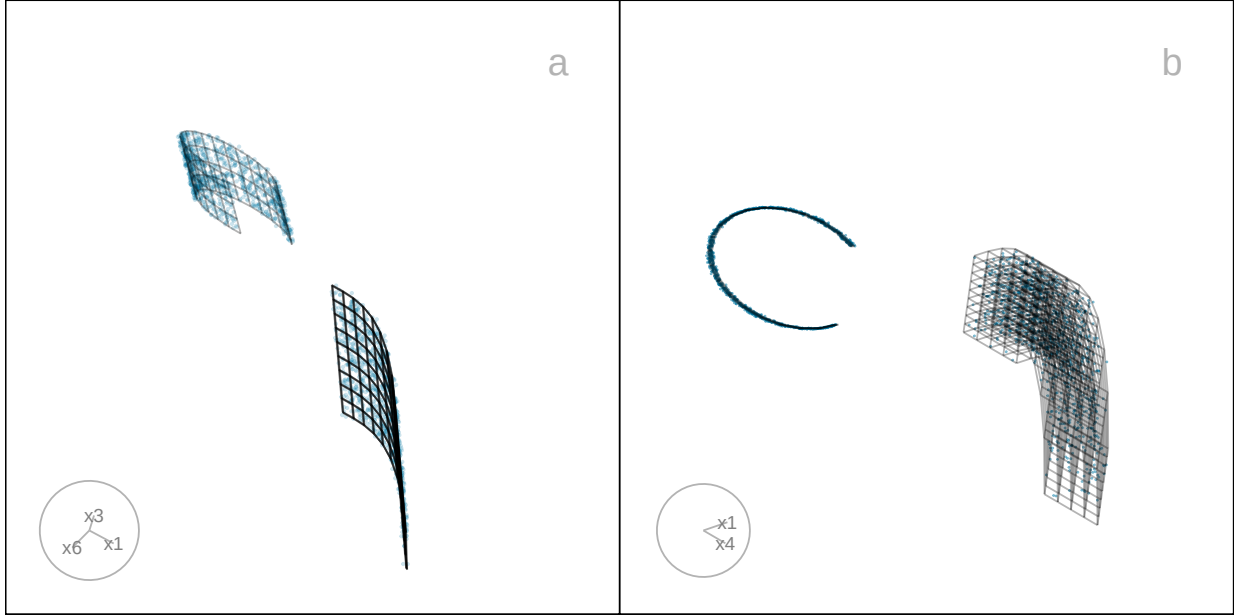


Figure 1: Two projections of the p - D true model overlaying the data are shown in a, b. Video of the langevitour animations is available at <https://youtu.be/35TrnYJsUUI>.

6 Computing hexagon grid configurations

Given range of embedding component, r_2 , number of bins along the x-axis, b_1 , and buffer proportion, q , hexagonal starting point coordinates, $s_1 = -q$, and $s_2 = -qr_2$. The purpose is to find width of the hexagon, a_1 and number of bins along the y-axis, b_2 .

Geometric arguments give rise to the following constraints.

$$\min a_1 \text{ s.t.}$$

$$s_1 - \frac{a_1}{2} < 0, \quad (1)$$

$$s_1 + (b_1 - 1) \times a_1 \geq 1, \quad (2)$$

$$s_2 - \frac{a_2}{2} < 0, \quad (3)$$

$$s_2 + (b_2 - 1) \times a_2 \geq r_2. \quad (4)$$

Since a_1 and a_2 are distances,

$$a_1, a_2 > 0.$$

Also, $(s_1, s_2) \in (-0.1, -0.05)$ as these are multiplicative offsets in the negative direction.

Equation 1 can be rearranged as,

$$a_1 > 2s_1$$

which given $s_1 < 0$ and $a_1 > 0$ will *always* be true. The same logic follows for Equation 3

and substituting $a_2 = \sqrt{3}a_1/2$, and $s_2 = -qr_2$ to Equation 3 can be written as,

$$a_1 > -\frac{4}{\sqrt{3}}qr_2$$

Also, substituting $a_2 = \sqrt{3}a_1/2$, $s_2 = -qr_2$ and rearranging Equation 4 gives:

$$a_1 \geq \frac{2(r_2 + qr_2)}{\sqrt{3}(b_2 - 1)}. \quad (5)$$

Similarly, substituting $s_1 = -q$ Equation 2 becomes,

$$a_1 \geq \frac{(1 + q)}{(b_1 - 1)}. \quad (6)$$

This is a linear optimization problem. Therefore, the optimal solution must occur on a vertex. So, by setting Equation 5 equals to Equation 6 gives,

$$\frac{2(r_2 + qr_2)}{\sqrt{3}(b_2 - 1)} = \frac{(1 + q)}{(b_1 - 1)}.$$

After rearranging this,

$$b_2 = 1 + \frac{2r_2(b_1 - 1)}{\sqrt{3}}$$

and since b_2 should be an integer,

$$b_2 = \left\lceil 1 + \frac{2r_2(b_1 - 1)}{\sqrt{3}} \right\rceil. \quad (7)$$

Furthermore, with known b_1 and b_2 , by considering Equation 2 or Equation 4 as the *binding* or *active constraint*, can compute a_1 .

If Equation 2 is active, then,

$$\frac{(1+q)}{(b_1-1)} < \frac{2(r_2+qr_2)}{\sqrt{3}(b_2-1)}.$$

Rearranging this gives,

$$r_2 > \frac{\sqrt{3}(b_2-1)}{2(b_1-1)}.$$

Therefore, if this equality is true, then

$$a_1 = \frac{(1+q)}{(b_1-1)},$$

otherwise,

$$a_1 = \frac{2r_2(1+q)}{\sqrt{3}(b_2-1)}.$$

7 Binning the data

Points are assigned to the bin they fall into based on the nearest centroid. If a point is equidistant from multiple centroids, it is assigned to the centroid with the smallest bin ID.

8 Area of a hexagon

The area of a hexagon is defined as $A = 3\sqrt{3}l^2/2$, where l is the side length of the hexagon.

l can be computed using a_1 and a_2 .

By applying the Pythagorean theorem, we obtain,

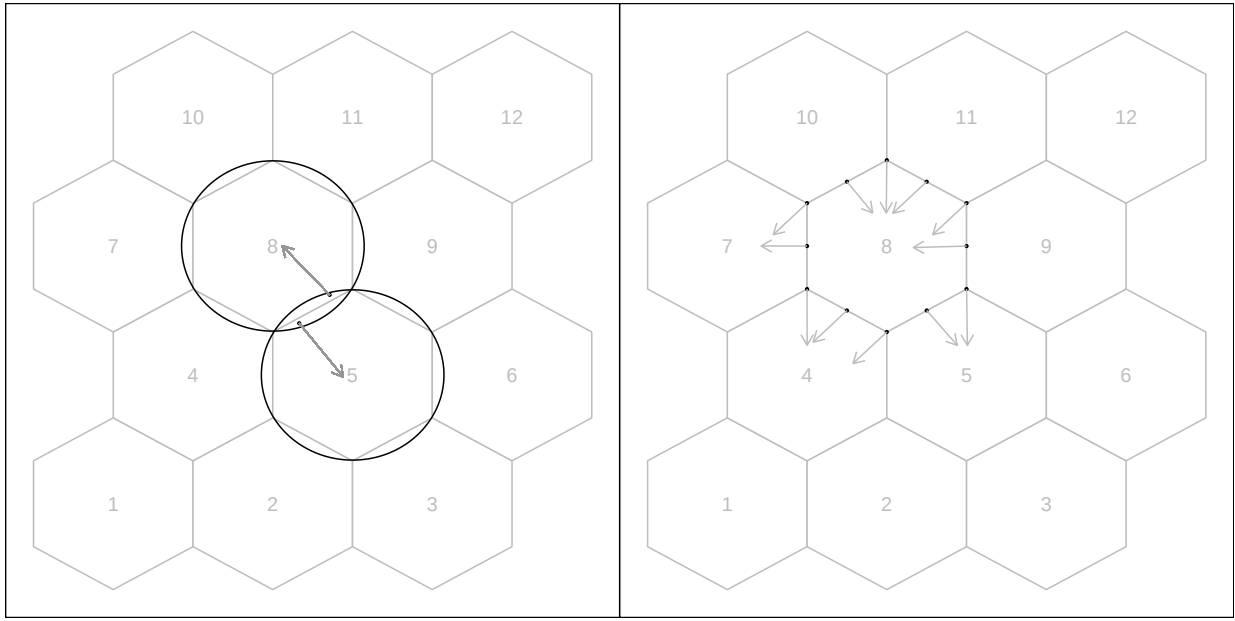


Figure 2: Binning the data. Points are assigned to the nearest centroid. If a point is equidistant from multiple centroids, assigned to the centroid with the smallest bin ID.

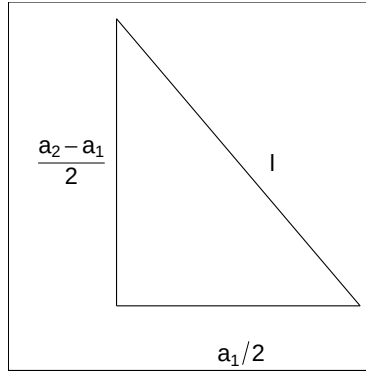


Figure 3: The components of the right triangle illustrating notation.

$$l^2 = \left(\frac{a_1}{2}\right)^2 + \left(\frac{a_2 - l}{2}\right)^2.$$

Next, rearranging the terms, we get,

$$l^2 - \left(\frac{a_2 - l}{2}\right)^2 = \left(\frac{a_1}{2}\right)^2,$$

$$\left[l - \left(\frac{a_2 - l}{2}\right)\right] \left[l + \left(\frac{a_2 - l}{2}\right)\right] = \left(\frac{a_1}{2}\right)^2,$$

$$3l^2 + 2a_2l - (a_1^2 + a_2^2) = 0.$$

Finally, by solving the quadratic equation, we compute,

$$l = \frac{-2a_2 \pm \sqrt{4a_2^2 - 24[-(a_1^2 + a_2^2)]}}{6},$$

$$l = \frac{-a_2 \pm \sqrt{a_2^2 - 6[-(a_1^2 + a_2^2)]}}{3},$$

where $l > 0$.

9 Tuning

The model fitting depends on several parameters, including the hexagon bin parameters and the low-count bin removal process. The hexagon bin parameters define the bottom-left bin position (s_1, s_2) , the number of bins in the horizontal direction (b_1), which determines the number of bins in the vertical direction (b_2), the total number of bins (b), and the number of non-empty bins (m). Low-count bins are removed using standardized bin counts, defined as $w_h = n_h/n$, for $h = 1, \dots, m$.

Default values are provided for these parameters. In particular, the default number of bins is determined from the sample size by setting $b_1 = n^{1/3}$, consistent with the Diaconis–

Freedman rule (Freedman & Diaconis 1981), with b_2 computed to preserve the aspect ratio of the layout.

Rather than selecting a single set of parameters a priori, we recommend a structured procedure to examine the stability of the fitted model across a range of values. A practical workflow is as follows:

1. **Select a grid of binwidths:** Choose a sequence of values for b_1 ranging from coarse to fine resolution (e.g., from small values up to $b_1 = \sqrt{n/r_2}$). This controls the level of detail in the fitted wireframe.
2. **Compute HBE across the grid:** For each choice of b_1 , compute the HBE. Stable regions where HBE changes slowly indicate parameter values where the fitted model is not overly sensitive to binning resolution.
3. **Assess bin occupancy:** Examine summary quantities such as average bin count ($\bar{n}$), average standardized bin count ($\bar{w}$), and the proportion of non-empty bins (m/b).

Reasonable parameter choices typically correspond to:

- sufficiently large $\bar{n}$ to avoid sparse bins, and
 - moderate values of m/b , indicating that bins cover the data without excessive fragmentation.
4. **Check robustness to bin removal:** Evaluate the effect of removing low-count bins using a range of thresholds. Parameter values where HBE remains stable under small amounts of bin removal indicate robustness of the fitted structure.
 5. **Compare across layouts (if applicable):** When comparing multiple NLDR layouts, examine HBE as a function of average bin count or standardized bin count. This

helps account for differences in point density between layouts.

This process does not yield a single “optimal” parameter choice, but instead identifies a range of stable and interpretable settings. The goal is to avoid parameter values that produce unstable fits, overly sparse binning, or misleading structure in the fitted wireframe.

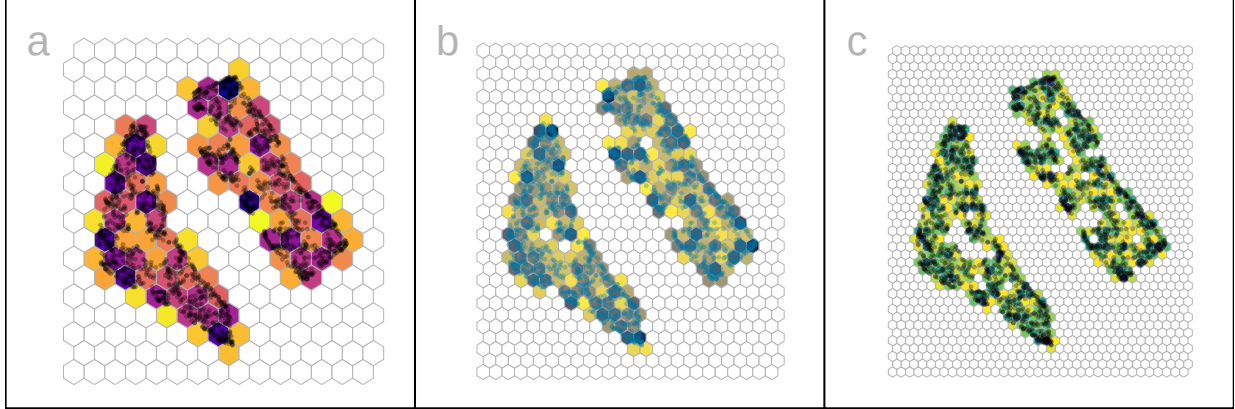


Figure 4: Hexbin density plots of tSNE layout of the 2NC7 data, using three different b_1 specifications yielding different b_2, b, m : (a) **15**, 18, 270, 98, (b) **24**, 29, 696, 209, and (c) **35**, 42, 1470, 386. Color indicates standardized counts, dark indicating high count, and light indicating low count. At the smallest binwidth, the data structure is discontinuous, suggesting that there are too many bins.

10 Curiosities about NLDR results discovered by examining the model in the data space

With the drawing of the model in the data, several interesting differences between NLDR methods can be observed.

10.1 Some methods appear to order points in the layout

The 2- D model representations generated from some NLDR methods, especially PaCMAP, are unreasonably flat or like a pancake. A simple example of this can be seen with data simulated to contain five 4- D Gaussian clusters. Each cluster is essentially a ball in 4- D , so

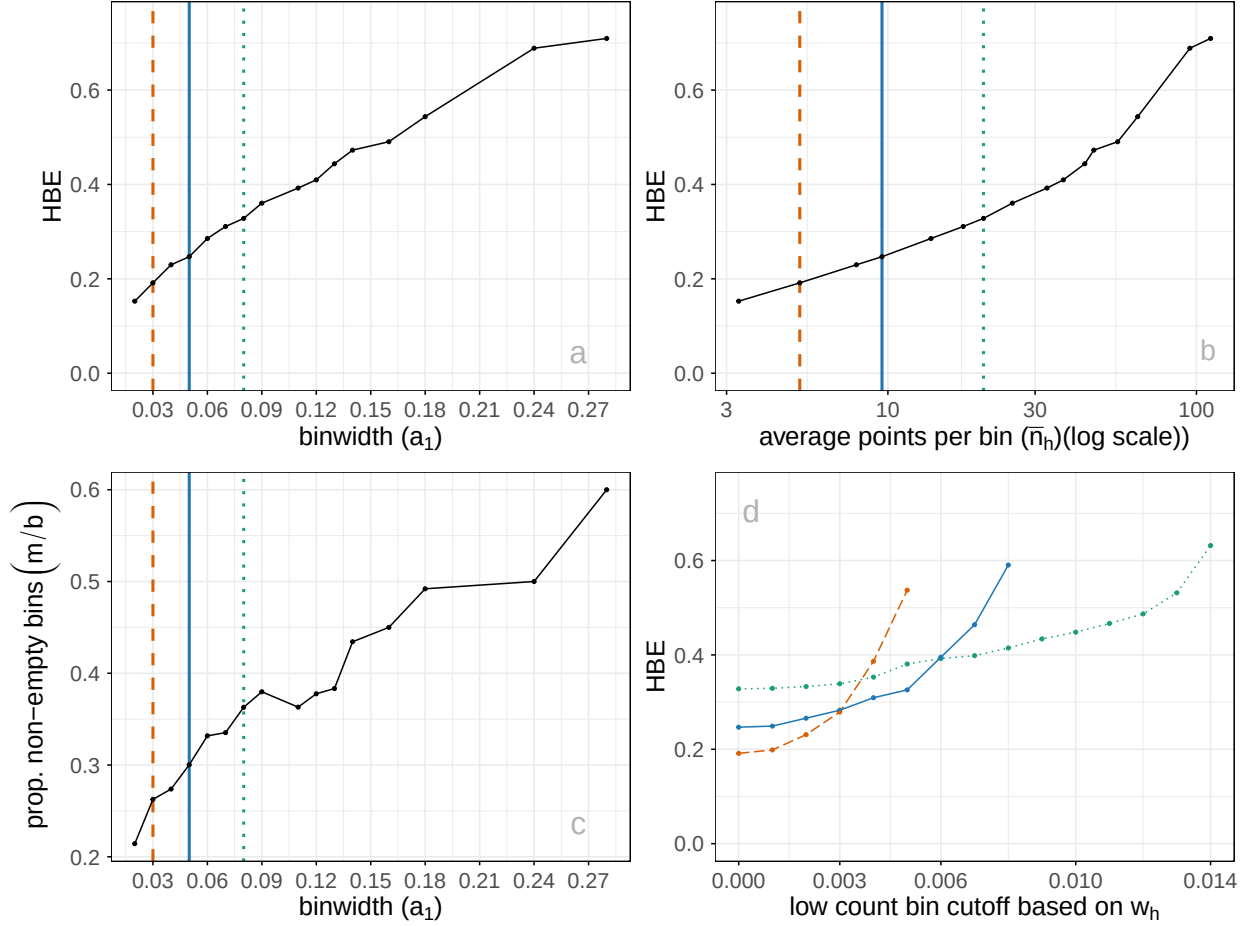


Figure 5: Various plots to help tune the model fit: (a) HBE vs a_1 , (b) HBE vs $\bar{n}_h$, (c) proportion of non-empty bins (m/b) vs a_1 , (d) HBE vs w_h cutoff used for removing low count bins. Color indicates the three binwidths show in Figure 6: 0.03 (orange dashed), 0.05 (blue solid), and 0.08 (green dotted). The better model fit will have low HBE, but reasonably sized bins that capture the data sufficiently. Proportion of non-empty bins tends to increase with a_1 (c). Removing a few low-count bins doesn't substantially change HBE, for all three binwidths (d).

there is no 2- D representation, rather the model in each cluster should resemble a crumpled sheet of paper that fills out 4- D .

Figure 6 a1, b1, c1 show the 2- D layouts for (a) tSNE, (b) UMAP, and (c) PaCMAP, respectively. The default hyper-parameters for each method are used. In each layout we can see an accurate representation where all five clusters are visible, although with varying degrees of separation.

The models are fitted to each these layouts. Figure 6 a2, b2, c2 show the fitted models in a projection of the 4- D space, taken from a tour. These clusters are fully 4- D in nature, so we would expect the model to be a *crumpled sheet* that stretches in all four dimensions. This is what is mostly observed for tSNE and UMAP. The curious detail is that the model for PaCMAP is closer to a *pancake* in shape in every cluster! This single projection only shows this in three of the five clusters but if we examine a different projection the other clusters exhibit the pancake also. While we don't know what exactly causes this, it is likely due to some ordering of points in the 2- D PaCMAP layout that induces the flat model. One could imagine that if the method used principal components on all the data, that it might induce some ordering that would produce the flat model. If this were the reason, the pancaking would be the same in all clusters, but it is not: The pancake is visible in some clusters in some projections but in other clusters it is visible in different projections. It might be due to some ordering by nearest neighbors in a cluster. The PaCMAP documentation doesn't provide any helpful clues. That this happens, though, makes the PaCMAP layout inadequate for representing the high-dimensional data.

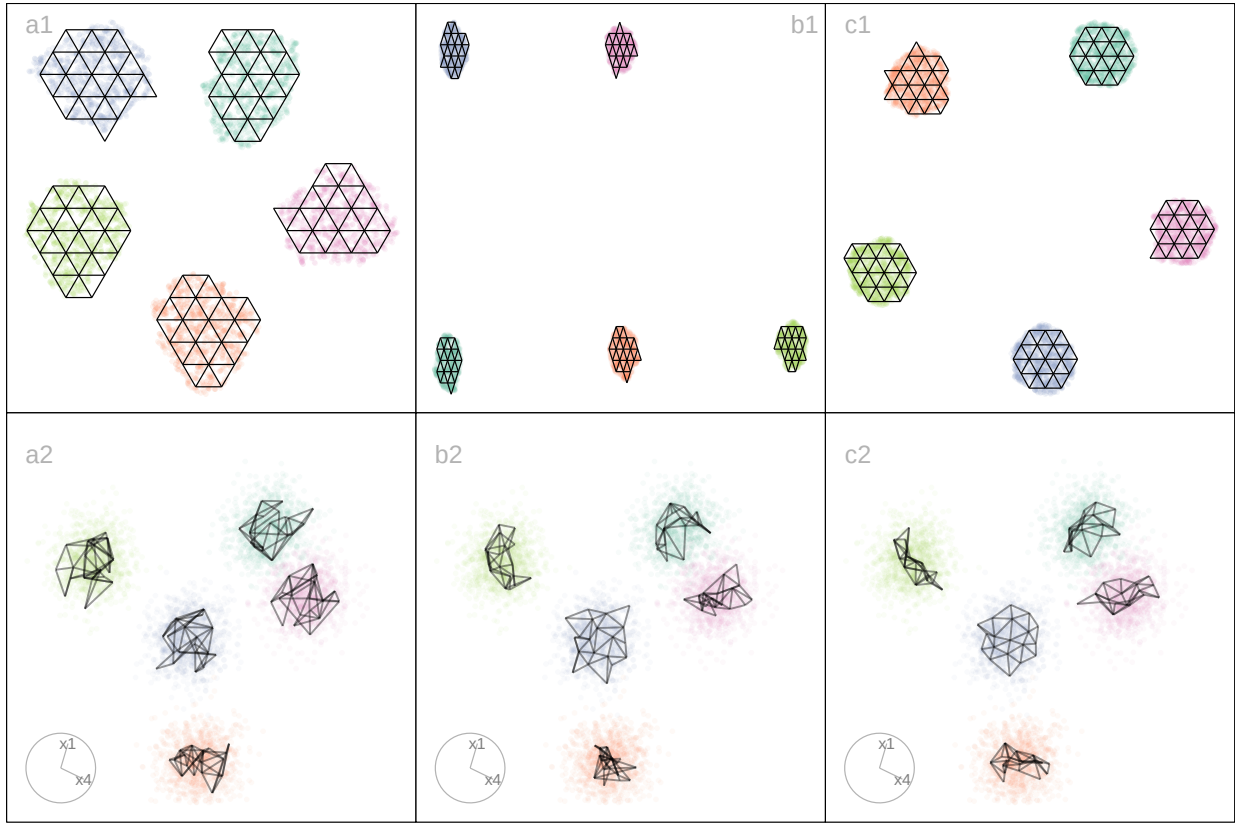


Figure 6: NLDR’s organize points in the 2- D layout in different ways, possibly misleadingly, illustrated using three layouts: (a) tSNE, (b) UMAP, (c) PaCMAP. The data has five Gaussian clusters in 4- D . The bottom row of plots shows a 2- D projection from a tour on 4- D revealing the differences generated by the layouts on the model fits. We would expect the model fit to be like that in (a2) where it is distinctly is separate for each cluster but like a hairball in each. This would indicate the distinct clusters, each being fully 4- D . With (c2), the curiosity is that the model is a 2- D pancake shape in 4- D , indicating that there is some ordering of points done by PaCMAP, possibly along some principal component axes. Videos of the langevitour animations are available at <https://youtu.be/I-kxCwVfqiQ>, <https://youtu.be/gD1P01FUPyU>, and https://youtu.be/MxJ_srOFQNk respectively.

10.2 Sparseness creates a contracted 2- D layout

Differences in density can arise by sampling at different rates in different subspaces of p - D . For example, the data shown in Figure 7 all lies on a 2- D curved sheet in 4- D , but one end of the sheet is sampled densely and the other very sparsely. It was simulated to illustrate the effect of the density difference on layout generated by an NLDR, illustrated using the tSNE results, but it happens with all methods.

Figure 7 (a2, b2) shows a 2- D layout for tSNE created using the default hyper-parameters. One would expect to see a rectangular shape if the curved sheet is flattened, but the layout is triangular. The other two displays show the residuals as a dot density plot (a1, b1), and a 2- D projection of the data and the model from 4- D (a3, b3). Using linked brushing between the plots, we can highlight points in the tSNE layout, and examine where they fall in the original 4- D . The darker (maroon) points indicate points that have been highlighted by linking. In row a, the points at the top of the triangle are highlighted, and we can see these correspond to higher residuals, and also to all points at the low density end of the curved sheet. In row b, points at the lower left side of the triangle are highlighted which corresponds to smaller residuals and one corner of the sheet at the high density end of the curved sheet.

The tSNE behaviour is to squeeze the low density area of the data together into the layout. This is common in other NLDR methods also, which means analysts need to be aware that if their data is not sampled relatively uniformly, apparent closeness in the 2- D may correspond to sparseness in p - D .

11 PBMC3k: comparison with results of scDEED recommendations

As we were writing this paper [Xia et al. \(2023\)](#) appeared proposing a new method called scDEED helping to assess the validity of a 2- D embedding. scDEED calculates a reliability score for each cell embedding based on the similarity between the cell’s 2- D embedding neighbors and its neighbors prior to embedding. A low reliability score suggests a dubious embedding. It can help in the deciding on optimal hyper-parameters. Here we illustrate

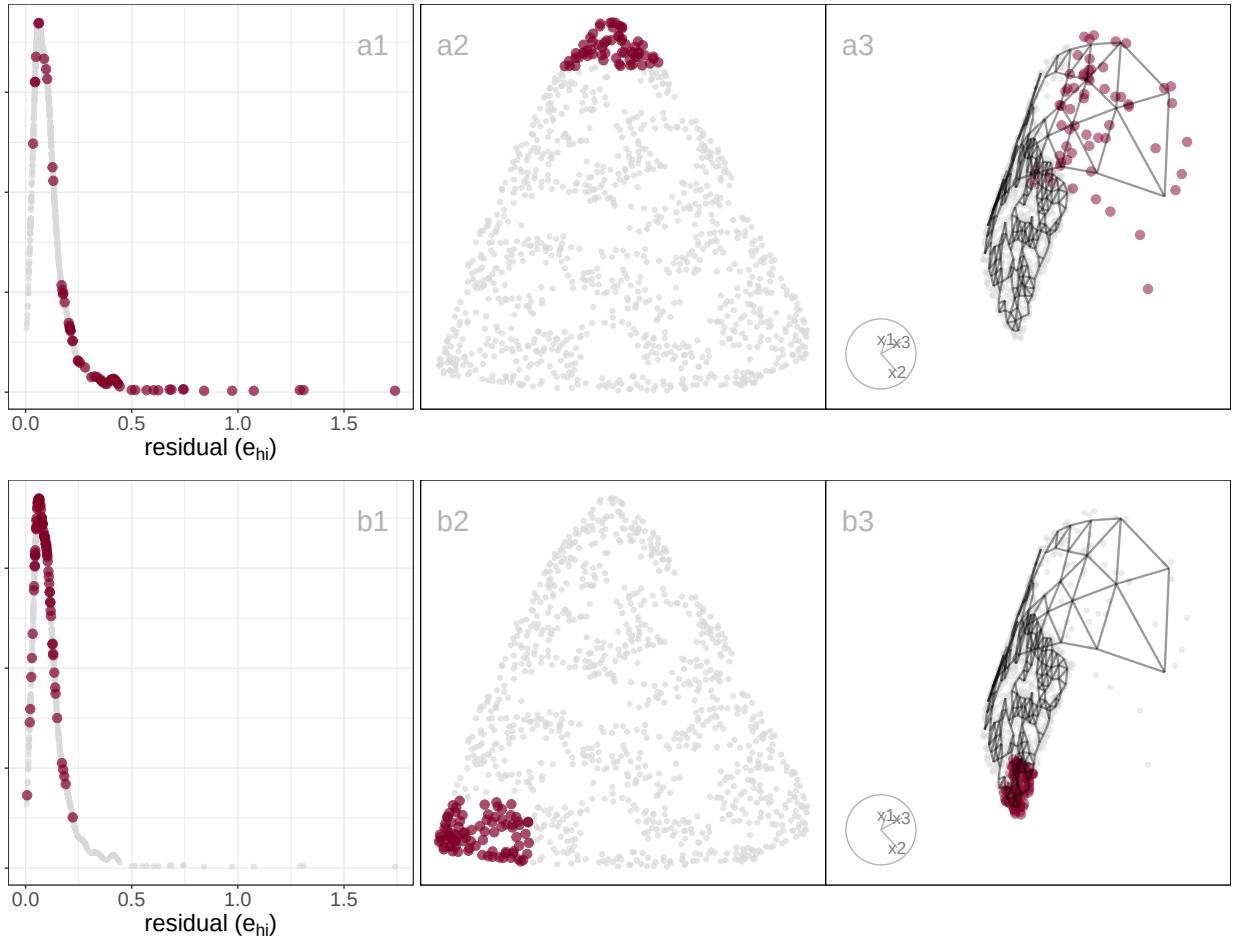


Figure 7: Exploring the effect of density on the NLDR layout using a 2- D curved sheet in 4- D with different density at each end. Three plots are linked: density plot of residuals (a1, b1), NLDR layout (a2, b2), projection of 4- D model and data (a3, b3). The brown points indicate the selected set, which are different in each row. In (a2), the top part of the triangular shape is selected which corresponds to higher residuals (a1) and the sparse end of the structure (a3). In (b2) one of other corners is highlighted, which can be seen to correspond to low residuals (b1) and one side of the dense end of the data (b3). While the tSNE layout represents the dense end of the sheet correctly as two corners in the layout, it contracts the sparse end of the sheet into a single corner. Video of the langevitour animation is available at <https://youtu.be/-KsQH0rII2A>.

how our method compares with the results from scDEED.

Note that [Xia et al. \(2023\)](#) uses a different PBMC dataset than that used by [Chen et al. \(2024\)](#), shown by us in the main paper example, which is why this comparison is shown here and not in the main paper. Their data contains 31,021 cells including cell type labels, and the gene expression levels were in the unit of log-transformed UMI count per 10,000. They

focused on three sequencing methods (inDrops, DropSeq, and SeqWell) and four common cell types Cytotoxic T cell, CD4+T cell, CD14+ Monocyte, and B cell. Pre-processing follows the process in [Xia et al. \(2023\)](#) again using the Human Peripheral Blood Mononuclear Cells (PBMC) data.

For illustration purposes, we only selected cells generated with inDrops ($n = 5858$ cells). Also, [Xia et al. \(2023\)](#) used first 9 principal components to generate the UMAP and tSNE with default hyper-parameters. The objective is to what scDEED suggests is the best layout with what HBE would choose. Layout a (Figure 8) is generated from the hyper-parameters suggested by [Chen et al. \(2024\)](#), and layout b (Figure 8) is with suggested hyper-parameters by scDEED to be more accurate. The HBE vs binwidth (a_1) plot (Figure 8) illustrates that our approach would suggest that scDEED is correct here, that layout b is more accurately reflecting the cluster structure in the PBMC data. This is also supported by examining the models in the data space as shown in Figure 9.

12 Compare HBE with existing evaluation metrics

Figure 12 and Figure 13 compare HBE with commonly used evaluation metrics such as rRTA, rARNX, rSC, and rGS across multiple NLDR layouts. These visual comparisons highlight that HBE behaves differently from these existing metrics due to the different settings involved.

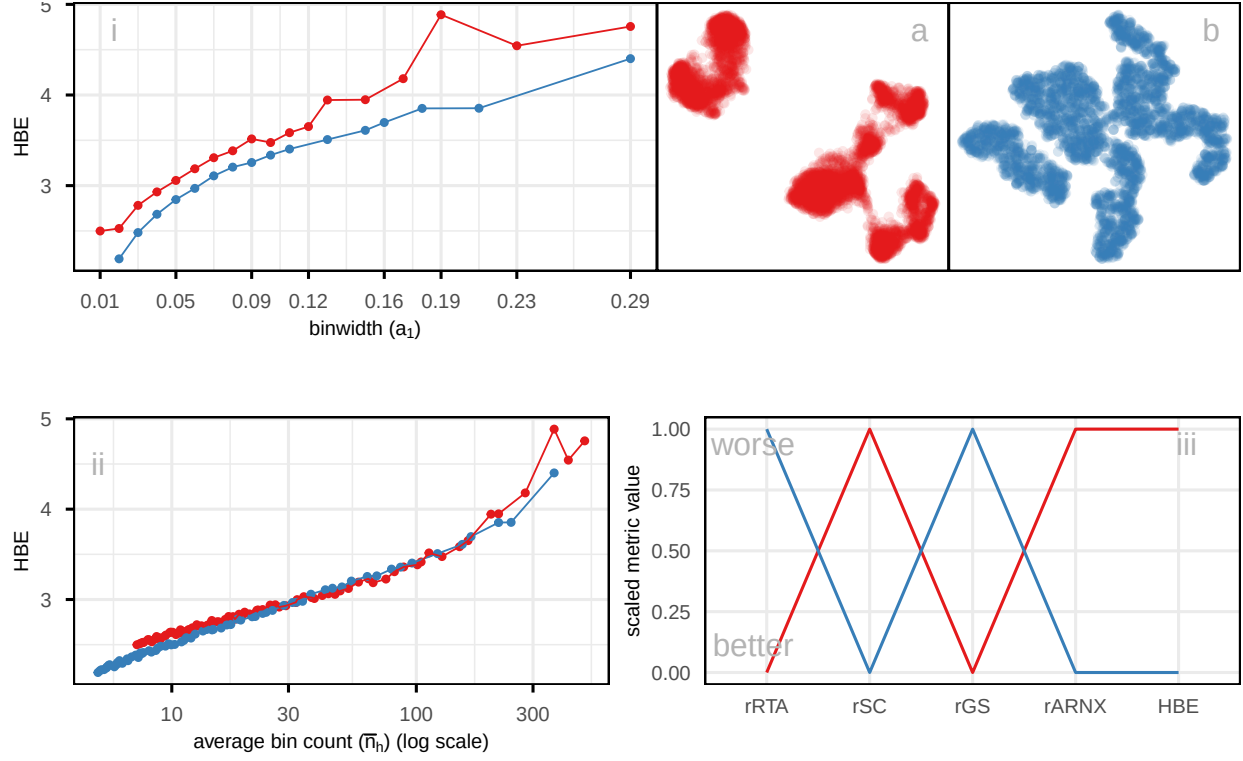


Figure 8: Comparing the published layout (a) with what would be suggested to be optimal by scDEED (b), using HBE for varying (i) binwidth (a_1), and (ii) average bin count ($\bar{n}_h$), on a subset of PBMC3k data. Color represents NLDR layouts. HBE would corroborate that the scDEED optimized layout is better than what was originally published. Plot (ii), which accounts for the density within clusters by using average bin count, shows reduced differences between layouts, indicating that part of the variation in (i) is driven by cluster density rather than true structural differences. Comparison of scaled evaluation metrics (iii) (rRTA, rSC, rGS, rARNX, and HBE using $a_1 = 0.04$) for two NLDR layouts of the PBMC3k data the originally published layout (a) and the scDEED optimized layout (b). Each line represents a layout, with color matching the corresponding scatterplots. Most metrics (rSC, rARNX, and HBE) consistently indicate that the optimized layout (b) provides a better representation, while rRTA, and rGS slightly favor the published layout.

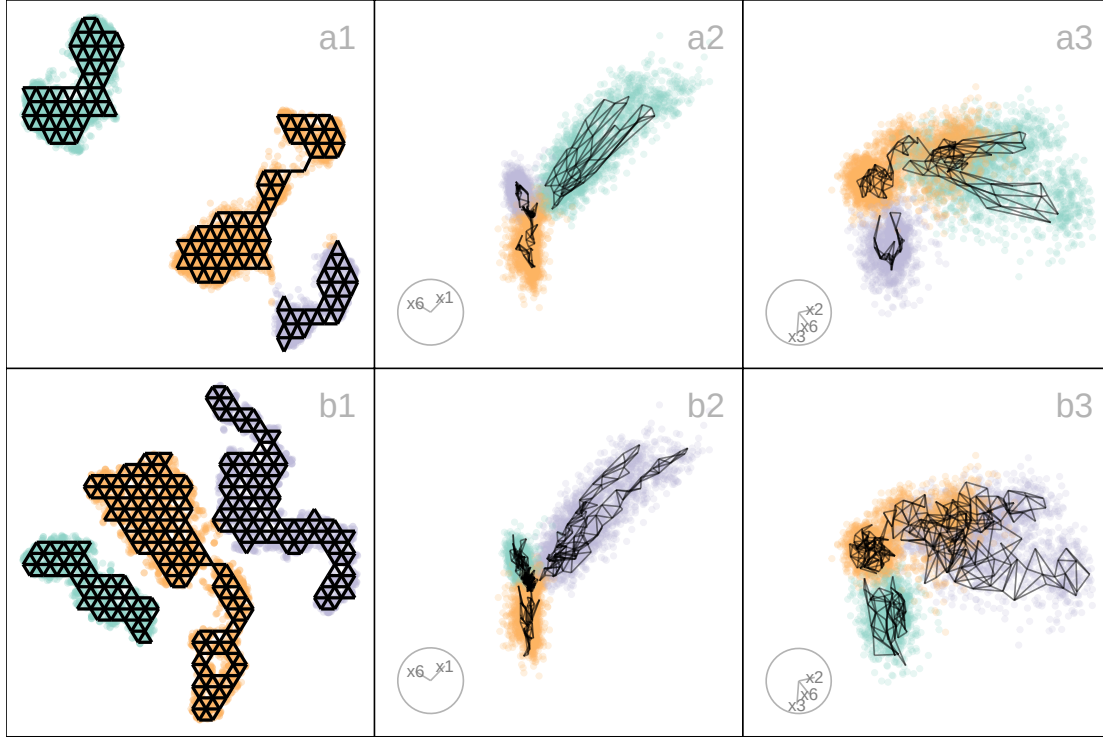


Figure 9: Compare the published 2- D layout (Figure 8 a) made with UMAP and the 2- D layout made with tSNE selected as optimal by scDEED (Figure 8 b) and also HBE (Figure 8). The two plots on the right show projections from a tour, with the models overlaid. The published layout a suggested three separated clusters with two of them are close, but this is not present in the data. While there may be three clusters they are not well-separated. The difference in model fit also indicates this: the published layout a does not capture the nonlinear structure of the clusters like the model generated from layout b. This supports the choice that layout b is the better representation of the data, because it shows close clusters. Videos of the langevitour animations are available at <https://youtu.be/ffiB4MGWyn8> and <https://youtu.be/e7XNL18co1c> respectively.

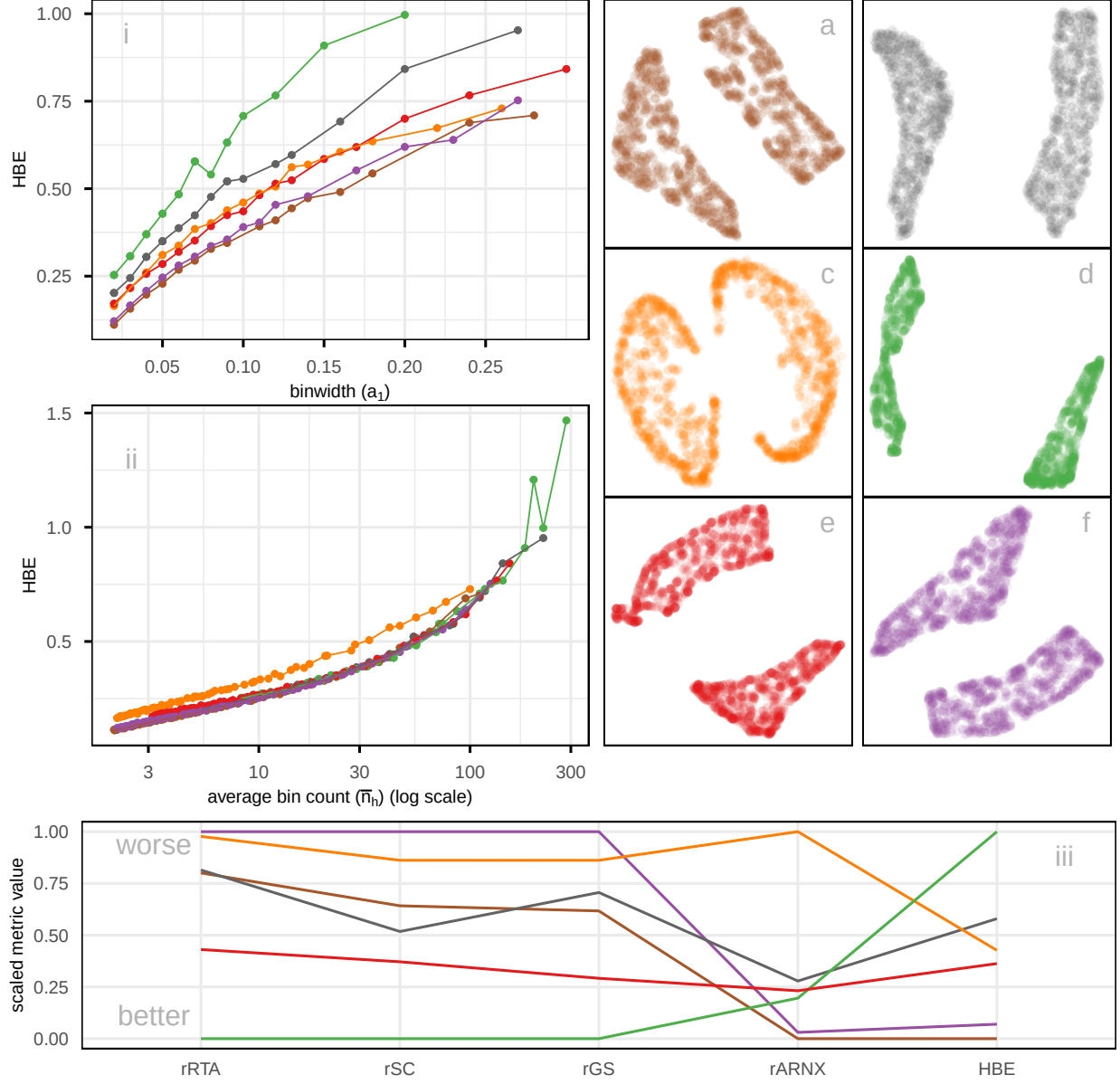


Figure 10: Assessing which of the 6 NLDR layouts (a-f) on the 2NC7 data is the better representation using HBE for varying (i) binwidth (a_1), and (ii) average bin count ($\bar{n}_h$). Color represents the NLDR layout. Layout d is universally poor. Layouts b and e that show two close clusters are universally suboptimal. Layout f with little separation performs well at tiny binwidth (where most points are in their own bin) and poorly as binwidth increases. Layout e has a small separation with oddly shaped clusters. Layout a is the best choice. Plot (ii), which compares HBE values with respect to average bin count, helps account for differences in cluster density; here, the variation among layouts is reduced, showing that some differences observed in (i) arise from density rather than true structure. Comparison of scaled evaluation metrics (rRTA, rSC, rGS, rARNX, and HBE using $a_1 = 0.05$) for the six NLDR layouts computed on the 2NC7 data using a parallel coordinate plot (iii). Color of the line indicates NLDR layout.

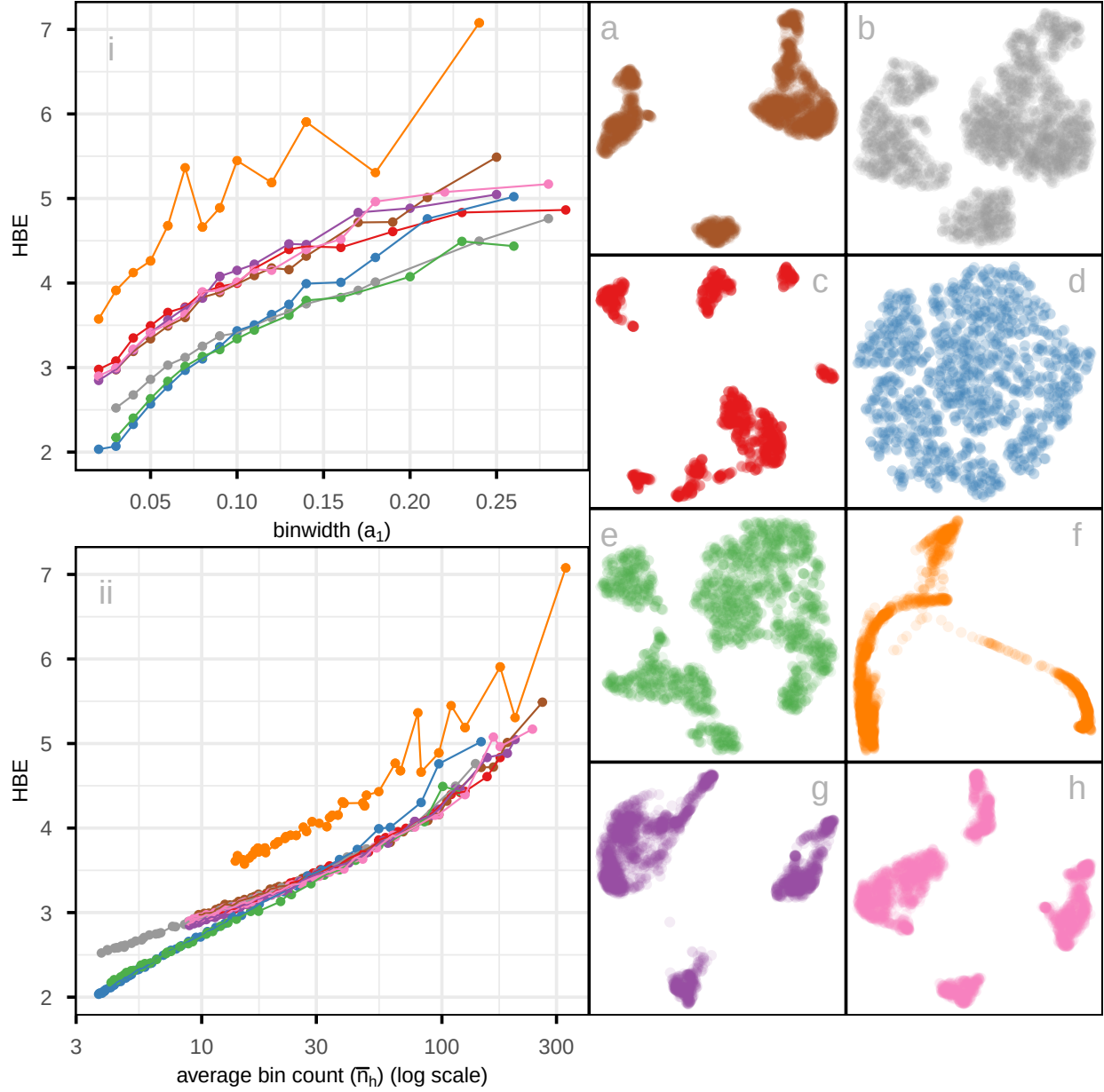


Figure 11: Assessing which of the 8 NLDR layouts on the PBMC3k data is the better representation using HBE for varying (i) binwidth (a_1), and (ii) average bin count ($\bar{n}_h$). Color used for the lines and points in the left plot and in the scatterplots represents NLDR layout (a-h). Layout f is universally poor. Layouts a, c, g, and h that show large separations between clusters are universally suboptimal. Layout d with little separation performs well at tiny binwidth (where most points are in their own bin) and poorly as binwidth increases. The choice of the best is between layouts b and e, which have small separations between oddly shaped clusters. Layout e is chosen as the best. Plot (ii), which accounts for the density within clusters by using average bin count, shows reduced differences between layouts, indicating that part of the variation in (i) is driven by cluster density rather than true structural differences.

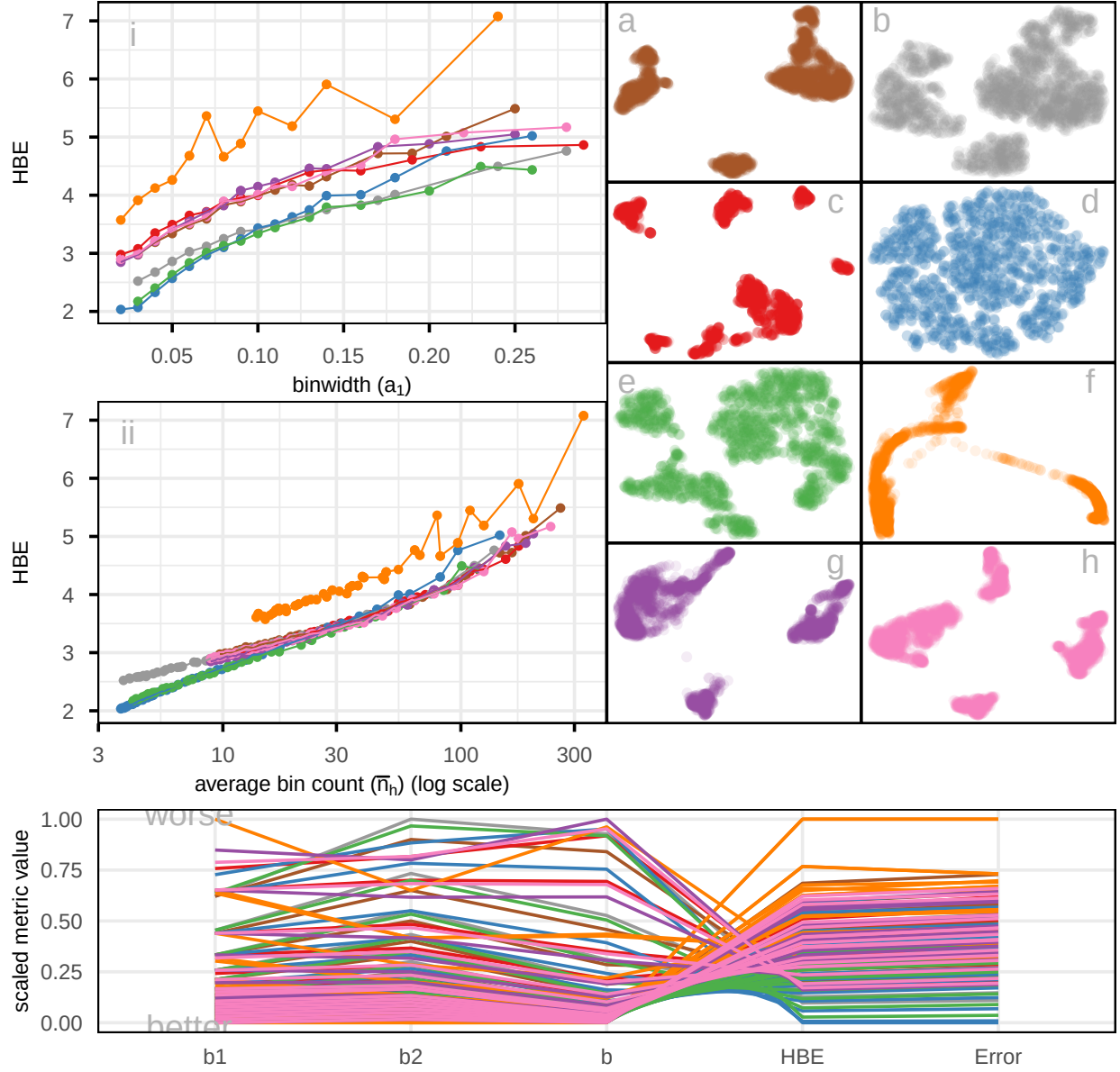


Figure 12: Comparison of scaled evaluation metrics (rRTA, rSC, rGS, rARNX, and HBE with $a_1 = 0.06$) for the eight NLDR layouts computed on the PBMC3k data, shown as a parallel coordinate plot. The color of each line corresponds to a NLDR layout. All, except rGS and rARNX agree that layout e is best or very close to best. Layout d is best according to HBE and rARNX but considered to be much less optimal by rRTA, rSC and rGS. Layout f is considered poor by rARNX and HBE. Layout a is considered close to the best by rGS and rSC.

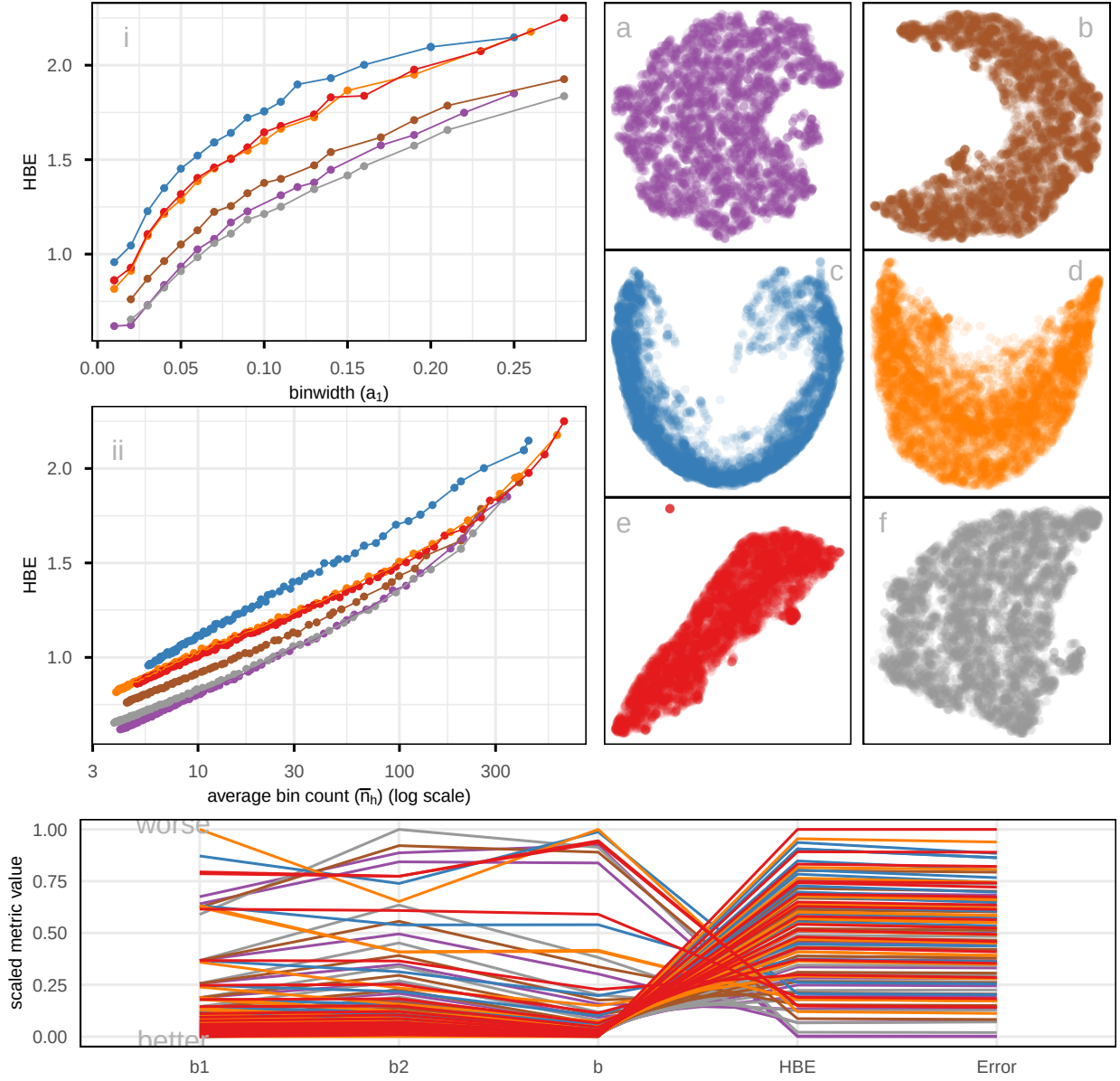


Figure 13: Comparison of scaled evaluation metrics (rARNX, rRTA, rSC, rGS, and HBE using $a_1 = 0.04$) for six NLDR layouts computed on the MNIST digit 1 data using a parallel coordinate plot. Each line represents a layout (a–f), with colors corresponding to the scatterplots shown on the right. The metrics display different ranking patterns, indicating that no single measure fully captures embedding quality. Layout a is identified as the best according to HBE and rRTA, but is considered much less optimal by rARNX, rSC, and rGS. Layout e is considered the worst, or close to the poorest, by all metrics. Layouts a and f show similar patterns of agreement across metrics, except for rRTA, where layout a performs the best and layout f the worst. Layout c is the worst in rARNX, rSC, and HBE.

References

- Chen, Z., Wang, C., Huang, S., Shi, Y. & Xi, R. (2024), ‘Directly Selecting Cell-type Marker Genes for Single-cell Clustering Analyses’, *Cell Reports Methods* **4**(7), 100810. <https://www.sciencedirect.com/science/article/pii/S2667237524001735>.
- Freedman, D. A. & Diaconis, P. (1981), ‘On the Histogram as a Density Estimator: L2 Theory’, *Probability Theory and Related Fields* **57**, 453–476. <https://doi.org/10.1007/BF01025868>.
- Xia, L., Lee, C. & Li, J. J. (2023), ‘scDEED: A Statistical Method for Detecting Dubious 2D Single-cell Embeddings’, *bioRxiv* . <https://www.biorxiv.org/content/early/2023/04/25/2023.04.21.537839>.